Advanced R Programming

Essential Packages and Ecosystem

Ethiopian Statistics Service (ESS)

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Training Roadmap

1. Introduction to R Ecosystem
2. The Tidyverse Ecosystem
3. High-Performance Data Wrangling
4. Machine Learning Frameworks
5. Statistical Modeling Packages
6. Package Distribution Sources
7. Training Program Structure
8. Key Takeaways

The Modern R Workflow

The R Ecosystem Landscape

The R ecosystem has evolved into a **cohesive framework** for professional data science, centered around the tidyverse philosophy.

- **5 Key Pillars** of Advanced R Programming:
  1. Data Wrangling & Manipulation
  2. Data Visualization
  3. Statistical Modeling
  4. Machine Learning
  5. Functional Programming
- **Core Philosophy**: "Tidy data" with consistent API design
- **Modern Toolset**: Packages that work seamlessly together

Training Objective

Master the essential packages that enable production-grade data analysis workflows.

The Tidyverse: Modern R Workflow

Core Tidyverse Packages

Package	Purpose
`ggplot2`	Advanced visualization
`dplyr`	Data manipulation
`tidyr`	Data reshaping
`lubridate`	Date-time handling
`purrr`	Functional programming
`stringr`	String operations

Key Philosophy

- Consistent syntax
- Pipe-friendly (%>%)
- Works with `tibbles`
- "One function, one task"

Powerful Pattern

```
data %>% filter(condition) %>% mutate(new = f(x)) %>%
ggplot(aes(...))
```

dplyr: Data Manipulation

Core Verbs

`filter()`	Row subsetting
`select()`	Column selection
`mutate()`	Create new variables
`summarise()`	Aggregate data
`arrange()`	Sort rows
`join_*()`	Merge datasets

Advanced Features

- `across()`: Apply function to multiple columns
- `rowwise()`: Row-wise operations
- `group_by()`: Grouped operations
- `pivot_*()`: Data reshaping

Efficiency Tip: `across()` enables powerful vectorized operations

ggplot2: Data Visualization Grammar

The Grammar of Graphics Components

1. **Data**: The dataset to visualize
2. **Aesthetics (aes)**: Map variables to visual properties
3. **Geoms**: Geometrical objects (points, lines, bars)
4. **Facets**: Subset data into panels
5. **Themes**: Control non-data elements
6. **Scales**: Control mapping from data to aesthetics

Layered Construction

```
ggplot(data, aes(x, y, color = group)) +
geom_point() +
geom_smooth(method = "lm") +
facet_wrap(~ category) +
theme_minimal()
```

data.table: Speed and Efficiency

Syntax Structure

```
DT[i, j, by]
```

- **i** Row filtering (WHERE)
- **j** Column operations (SELECT)
- **by** Grouping (GROUP BY)

Key Advantages

- **Memory efficient**: Modification by reference
- **Super fast**: Optimized C code
- **No copying**: Avoids memory bloat
- **Handles GB-scale data**

Real-World Performance

Operation	`data.table` Speed
Grouped operations	10-100x faster
Merging datasets	2-5x faster
Large data loading	1.5-3x faster

data.table: Key Operations

Essential Functions

`setkey()`	Set primary key for fast merging
`merge()`	Fast joins (2-5x faster than base R)
`.SD`	Subset of data for grouped operations
`.N`	Count rows within groups
`shift()`	Lag/lead operations

Common Pattern

```
DT[condition, .(mean_x = mean(x), count = .N), by = group]
DT[order(-date), .(last_value = value[1]), by = id]
```

tidymodels: Modern ML Framework

Core Components

Package	Purpose
`parsnip`	Unified model specification
`recipes`	Feature preprocessing pipelines
`tune`	Hyperparameter tuning
`workflows`	Combine preproc + model
`yardstick`	Performance metrics
`rsample`	Resampling strategies

Workflow Construction

```
rf_spec <- rand_forest(mtry = tune()) %>%
set_engine("ranger") %>%
```

mlr3: Advanced ML Framework

Object-Oriented ML

`Learner`	Machine learning algorithm
`Task`	Data + target + features
`Resampling`	Cross-validation strategies
`Measure`	Performance metrics
`Benchmark`	Compare multiple learners

Key Features

- Extremely extensible
- Built-in parallelization
- Comprehensive visualization
- 100+ built-in learners

Advanced Use Case

```
benchmark_grid <- benchmark(
design = expand.grid(
learner =
list(lrn("classif.rpart"),
lrn("classif.svm")),
task = list(tsk("iris"),
tsk("sonar")))
```

Hmisc: Advanced Statistical Analysis

Core Statistical Functions

`rcorr()`	Correlation matrix with p-values
`describe()`	Comprehensive summaries
`cut2()`	Create categorical variables
`Ecdf()`	Empirical CDFs
`impute()`	Missing data handling

Advanced Visualization

- `histboxp()`: Boxplot + histogram
- `plot()`: Extended plotting methods
- `latex()`: Table generation
- `summary.formula()`: Formula-based summaries

Key Strength

Excellent for **clinical research** and **biostatistics** applications.

lessR: Simplified Statistical Reporting

Less Code, More Results

Each function replaces multiple standard R functions.

Key Functions

`Read()`	Import with automatic summaries
`CountAll()`	Tables with visualizations
`Regression()`	Comprehensive regression output
`ANOVA()`	Complete ANOVA with plots

Integrated Output

- Summary statistics
- Histograms
- Bar charts
- Violin/Box/Scatter plots
- Model diagnostics

Single-Call Analysis

```
Read("data.csv") %>% ANOVA(Response ~ Group, data = .)
```

R Package Distribution Ecosystem

Three Main Sources

Source	Purpose
CRAN	General p pose
Bioconductor	Bioinformati
GitHub	Development versions

Installation Methods

```
CRAN:
        install.packages("package"
Bioconductor:
        BiocManager::install("pack
GitHub:
        devtools::install_github("
```

Strategic Considerations

- CRAN: Stable, tested, recommended for production
- Bioconductor: Domain-specific, rigorous peer review
- GitHub: Latest features, experimental, use with caution

5-Day Advanced R Training Program

Day 1: Tidyverse Fundamentals

- `ggplot2` visualization grammar
- `dplyr` data manipulation workflow
- Piping and functional programming
- Real-world data cleaning challenges

Day 2: Advanced Data Manipulation

- `data.table` syntax and performance
- Large dataset handling strategies
- Modification by reference
- Real-world data wrangling

Day 3: Machine Learning Pipelines

- `tidymodels` or `mlr3` framework
- Model training and validation

5-Day Advanced R Training Program (cont.)

Day 4: Statistical Modeling & Reporting

- `Hmisc` for advanced analysis
- `lessR` for efficient reporting
- Interactive visualization with `plotly`
- Reproducible reporting with R Markdown

Day 5: Integration & Package Development

- Building R packages with `devtools`
- Version control with Git/GitHub
- Code testing with `testthat`
- Production-ready code practices
- Documentation with `roxygen2`

Capstone Project: End-to-end data analysis pipeline

Essential Learning Outcomes

Master the Ecosystem

1. **Tidyverse** for daily workflow
2. **data.table** for big data
3. **tidymodels/mlr3** for ML
4. **Hmisc/lessR** for stats
5. Package management strategies

Professional Skills

1. Efficient data manipulation
2. Reproducible analysis
3. Machine learning pipelines
4. Statistical reporting
5. Package development

Learn the tools. Master the workflow.

Resources & Next Steps

Official Documentation

- Tidyverse: www.tidyverse.org
- data.table: rdatatable.gitlab.io
- tidymodels: www.tidymodels.org
- mlr3: mlr3.mlr-org.com

Additional Learning

- **Books**: "R for Data Science" (Wickham), "Advanced R" (Wickham)
- **Cheatsheets**: RStudio's official cheatsheets
- **Community**: RStudio Community, Stack Overflow, R-bloggers

Ready to elevate your R skills!